Name: _____ Class: _____

JURONG PIONEER JUNIOR COLLEGE
JC2 Theory Practice Set 1 (2026)

COMPUTING **9569/01**

Higher 2

Paper 1 (Written) 3 hours

Additional materials: Answer Booklet

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.

Approved calculators are allowed.

You are reminded of the need for clear presentation in your answers.

Answer papers will be provided with the question paper.

Write your name and civics class on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Write your answers to every question on a fresh page of paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

This document consists of 9 printed pages.

1 CycleGo operates a bicycle-sharing service in Singapore. When a user scans the QR code on a bicycle, the CycleGo app decides whether the bicycle can be unlocked.

The app checks three conditions:

- whether the user's account has an unpaid penalty
- whether the user has completed identity verification
- whether the user's wallet balance is at least $5.00.

The rules are as follows:

- If the account has an unpaid penalty, the app displays "Account suspended".
- If there is no unpaid penalty but the user has not completed identity verification, the app displays "Verify identity".
- If there is no unpaid penalty, the user has completed identity verification and the wallet balance is at least $5.00, the bicycle is unlocked and the app displays "Bicycle unlocked".
- If there is no unpaid penalty, the user has completed identity verification but the wallet balance is less than $5.00, the app displays "Top up required".

(a) Copy and complete the decision table. Include all possible combinations and do not remove redundancies at this point.

Conditions	account has an unpaid penalty	Y	Y	Y	Y	N	N	N	N
	identity verification completed	Y	Y	N	N	Y	Y	N	N
	wallet balance is at least $5.00	Y	N	Y	N	Y	N	Y	N
Actions	Account suspended								
	Verify identity								
	Top up required								
	Bicycle unlocked								

[4]

(b) Draw a simplified decision table by removing redundancies from your answer in part (a). [2]

(c) Write pseudo-code to input the values for the three conditions and output the appropriate message. Use the identifiers in the table below.

Identifier	**Data type**	**Description**
`HasPenalty`	BOOLEAN	TRUE if the account has an unpaid penalty
`IsVerified`	BOOLEAN	TRUE if identity verification has been completed
`Balance`	REAL	the wallet balance, in dollars

[4]

(d) Users top up their wallets in whole dollars. Each top-up, `TopUp`, must be from $10 to $100 inclusive.

Copy and complete the table to give a suitable value of `TopUp` for each type of test data, together with a reason for your choice.

Type of test data	**Value of TopUp**	**Reason**
Normal		
Extreme		
Abnormal		

[3]

2 The recursive function `Convert` is used in a program that displays numbers in different number bases.

```
01  FUNCTION Convert(N : INTEGER, B : INTEGER) RETURNS STRING
02     DECLARE Symbols : STRING
03     Symbols ← "0123456789ABCDEF"
04     IF N < B THEN
05        RETURN Symbols[N]
06     ELSE
07        RETURN Convert(N DIV B, B) & Symbols[N MOD B]
08     ENDIF
09  ENDFUNCTION
```

- `N` is a positive integer and `B` is an integer from 2 to 16 inclusive.
- `Symbols[i]` returns the character at position `i` of the string `Symbols`. The first character is at position 0.
- `&` joins two strings together.
- `DIV` returns the integer quotient and `MOD` returns the remainder of an integer division.

(a) Draw a trace tree, or otherwise, to show the function calls made when `Convert(300, 16)` is executed. Show the parameters of each call and the value returned by each call. [3]

(b) (i) State the purpose of the function `Convert`. [1]

The Chinese character 新 has the Unicode code point 26032 in denary.

(ii) State the value returned by `Convert(26032, 16)`. [1]

(iii) Explain why this character cannot be represented using ASCII. [1]

(c) Explain how a stack is used when `Convert(300, 16)` is executed. [2]

(d) A programmer accidentally calls `Convert(300, 1)`. Describe what will happen when this call is executed and identify the type of error that occurs. [2]

(e) Write pseudo-code for an iterative function `ConvertIter(N, B)` that returns the same value as `Convert(N, B)` for all valid values of `N` and `B`. [4]

3 A music app stores a playlist as a linear linked list. The linked list uses dynamic memory allocation: each song is stored in a node that is created when the song is added and released when the song is removed.

Each node is an object with two attributes:

Attribute	Description
`Title`	the title of the song (STRING)
`Next`	a pointer to the next node, or NULL if there is no next node

The pointer `Head` points to the first node in the playlist, or is NULL if the playlist is empty. The songs are kept in the order in which they are to be played, which is not alphabetical order.

(a) Explain one advantage and one disadvantage of using dynamic memory allocation for the playlist instead of storing the songs in a fixed-size array. [2]

The playlist currently contains four songs:

```
Head → "Aurora" → "Breeze" → "Echo" → "Halo" → NULL
```

(b) The following operations are carried out in the order shown:

1. Insert "Drift" immediately after "Breeze"
2. Delete "Aurora"
3. Delete "Halo"

For each operation, describe the pointer changes that are made. Draw the final linked list. [3]

(c) Write pseudo-code for the procedure `DeleteTitle(Target : STRING)` that deletes the first node whose `Title` is `Target`. The procedure should output a suitable message if the playlist is empty or if `Target` is not found. [5]

(d) The playlist is later sorted into alphabetical order of title. Explain why a binary search is still not efficient for finding a song in this linked list. [2]

4 A library uses a hash table to look up books quickly by a 4-character book code, for example "AB12". The hash table, `HashTable`, is a one-dimensional array of 11 strings, indexed 0 to 10. An empty element contains the empty string "".

The hash function is shown below. `ASC(ch)` returns the ASCII code of the character `ch`, and `Code[i]` returns the character at position `i` of `Code`, where the first character is at position 0.

```
FUNCTION Hash(Code : STRING) RETURNS INTEGER
   DECLARE Total, i : INTEGER
   Total ← 0
   FOR i ← 0 TO LENGTH(Code) - 1
      Total ← Total + ASC(Code[i])
   NEXT i
   RETURN Total MOD 11
ENDFUNCTION
```

The ASCII codes for "A" to "Z" are 65 to 90, and the ASCII codes for "0" to "9" are 48 to 57.

A collision is resolved by linear probing: if an element is occupied, the next element is tried, wrapping around from index 10 to index 0.

(a) Show that `Hash("AB12")` returns 10. [2]

(b) The following book codes are inserted, in the order given, into an empty hash table. The value returned by `Hash` for each code is shown.

Book code	AB12	CS21	BA21	MX07	KQ30	ZE15	PD44	HT09
Hash value	10	7	10	4	2	8	10	8

Copy and complete the table to show the final contents of `HashTable`.

Index	0	1	2	3	4	5	6	7	8	9	10
Book code											

[3]

(c) Write pseudo-code for the function `FindCode(Code : STRING) RETURNS INTEGER` that returns the index of `Code` in `HashTable`, or returns -1 if `Code` is not in the table. Your function must use `Hash` and must stop searching as soon as it can be certain that `Code` is not in the table. [4]

(d) With reference to the book codes in part (b), explain why this hash function is likely to cause many collisions. [1]

5 A data analytics company processes records stored in arrays. The following values are stored in the array `Data`:

```
38   14   52   7   29   41   3   19
```

(a) A bubble sort is used to sort `Data` into ascending order. State the contents of `Data` after each of the first two passes of the bubble sort. [2]

(b) Use annotated diagrams to show how a merge sort algorithm will sort the original contents of `Data` into ascending order. [3]

(c) Copy and complete the table to state the worst-case time complexity of each algorithm, using Big-O notation.

Algorithm	**Worst-case time complexity**
Bubble sort	
Insertion sort	
Quicksort	
Merge sort	

[2]

(d) Every night, the company must sort about 2 million transaction records within a fixed time window. Explain why merge sort may be chosen instead of quicksort for this task. [2]

After sorting, the contents of `Data` are:

Index	0	1	2	3	4	5	6	7
Value	3	7	14	19	29	38	41	52

A binary search calculates the middle index using `Mid ← (Low + High) DIV 2`. Initially, `Low` is 0 and `High` is 7.

(e) State, in order, the values in `Data` that are compared with the target when a binary search is carried out for the value 40. State the outcome of the search. [2]

(f) State the maximum number of elements that need to be examined when a binary search is carried out on a sorted array of 1000 elements. [1]

6 FitTrack is a fitness app that records workout activities. The app is to be implemented using object-oriented programming (OOP).

For every activity, the app stores:

- a unique activity ID
- the date of the activity
- the duration of the activity in minutes, which must be an integer from 1 to 600 inclusive.

There are three types of activity: running, swimming and cycling.

- For a run, the app also stores the distance in kilometres.
- For a swim, the app also stores the number of laps and the length of the pool in metres.
- For a cycle ride, the app also stores the distance in kilometres and the total elevation gain in metres.

Every activity has a method `CaloriesBurned()` that returns the estimated number of calories used. The calculation is different for each type of activity:

- run: 62 × distance
- swim: 0.12 × number of laps × length of pool
- cycle ride: 25 × distance + 0.4 × elevation gain.

Every activity also has a method `Summary()` that returns a string containing the activity ID, the date and the duration.

(a) Draw a class diagram for the FitTrack app. Your diagram must show:

- the superclass
- the subclasses
- the inheritance relationships
- the attributes, with suitable data types
- appropriate methods, including any methods that are overridden.

Indicate clearly which attributes and methods are private and which are public. [6]

(b) A list, `Workouts`, contains a mixture of run, swim and cycle ride objects. The app calculates the total number of calories used in all the workouts by calling `CaloriesBurned()` for every object in `Workouts`.

Explain how polymorphism is demonstrated in this calculation, and state one benefit of using polymorphism here. [2]

(c) Explain how encapsulation can be used to ensure that the duration of an activity is always from 1 to 600 inclusive. [2]

(d) State what is meant by instantiation, with reference to the FitTrack app. [1]

7 PawsCare is a veterinary clinic. It records appointments for pets and the medication prescribed at each appointment.

- Each owner has an owner ID, a name and a phone number. An owner may have one or more pets.
- Each pet has a pet ID, a name and a species, and belongs to one owner.
- Each appointment has an appointment ID and a date, and is for one pet.
- Each medication has a medication code, a name and a unit price.
- One or more medications may be prescribed at an appointment. The quantity of each medication prescribed at an appointment is recorded.

At present, all the data is stored in a single table, `ClinicVisit`, with the following table description. The primary key is underlined.

ClinicVisit(ApptID, ApptDate, PetID, PetName, Species, OwnerID, OwnerName, OwnerPhone, MedCode, MedName, UnitPrice, Quantity)

Some of the data in `ClinicVisit` is shown below. Some columns are not shown.

ApptID	ApptDate	PetID	PetName	OwnerID	OwnerName	MedCode	MedName	UnitPrice	Quantity
A101	2026-10-05	P01	Mochi	O11	Tan Wei Ling	M03	Deworm tablet	12.50	2
A101	2026-10-05	P01	Mochi	O11	Tan Wei Ling	M07	Ear drops	18.00	1
A102	2026-10-05	P02	Rex	O12	Ahmad Ismail	M03	Deworm tablet	12.50	1
A103	2026-10-12	P03	Bubbles	O11	Tan Wei Ling	M10	Vitamin paste	9.80	3

(a) Explain why `ClinicVisit` is not in second normal form (2NF). Refer to the data in your answer. [2]

(b) A table description can be expressed as:

TableName(Attribute1, Attribute2, Attribute3, ...)

The primary key is indicated by underlining one or more attributes. Foreign keys are indicated by using a dashed underline.

The data is to be stored in five tables in third normal form (3NF): `Owner`, `Pet`, `Appointment`, `Medication` and `Prescription`. Write table descriptions for the five tables. [4]

(c) Draw an entity-relationship (ER) diagram to show the five tables and the relationships between them. [3]

(d) Write an SQL query to display the name of each owner and the total cost of all the medication prescribed for that owner's pets. The total cost of each medication prescribed is its quantity multiplied by its unit price. The total cost for each owner should be labelled `TotalCost`.

Only include owners whose `TotalCost` is more than $50, and display the results in descending order of `TotalCost`. [3]

(e) PawsCare must keep the records of appointments from previous years, although these records are rarely used.

Explain the difference between a backup and an archive, with reference to the data stored by PawsCare. [2]

(f) PawsCare plans to store X-ray images, laboratory results and free-text consultation notes for each pet. The kinds of data stored differ greatly from one pet to another.

State one reason why a NoSQL database may be more suitable than a relational database for storing this data. [1]

8 QueueNow is a queue management system used at a polyclinic.

- At the entrance, patients register at self-service kiosks. Each kiosk is a client that connects to a queue server on the polyclinic's local area network (LAN).
- Patients can view their queue status on their mobile phones using a web application. The data is sent from the polyclinic's web server to the phones over the internet, which is a packet-switching network.

(a) Explain how a web page showing the queue status is transmitted from the polyclinic's web server to a patient's mobile phone over the internet. [2]

(b) The queue server is an iterative server that is implemented using socket programming.

Describe the steps carried out by the queue server so that it can receive a registration request from a kiosk and send back a queue number. [3]

(c) Each queue number message sent by the queue server to a kiosk is digitally signed.

Explain how the kiosk uses the digital signature to check that the message was sent by the queue server and has not been altered. [3]

(d) The polyclinic plans to use an Artificial Intelligence (AI) system to decide the queue priority of each patient, based on the symptoms that patients enter at the kiosks.

Discuss the social and ethical issues that may arise from this plan. [5]

END OF PAPER